

BEAUVECHAIN AB, Belgium

WMO: 064580

Lat: 50.7456N

Lon: 4.7633E

Elev: 370

StdP: 14.50

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 20.5	(c) 24.2	(d) 14.4	(e) 11.6	(f) 23.2	(g) 18.8	(h) 14.3	(i) 26.4	(j) 31.5	(k) 48.5	(l) 28.2	(m) 47.9	(n) 6.7	(o) 60	(p) 0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.7	(c) 85.0	(d) 67.5	(e) 81.0	(f) 66.1	(g) 77.7	(h) 64.6	(i) 69.4	(j) 80.7	(k) 67.5	(l) 77.7	(m) 66.0	(n) 75.3	(o) 6.9	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.6	95.9	73.5	63.9	90.1	71.6	62.4	85.5	70.2	33.6	80.9	32.1	78.1	30.9	75.6	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.4	(b) 21.1	(c) 18.5	(e) 16.4	(f) 91.8	(g) 5.6	(h) 3.0	(i) 12.3	(j) 93.9	(k) 9.0	(l) 95.6	(m) 5.9	(n) 97.3	(o) 1.8	(p) 99.4		
(a) 24.4	(b) 21.1	(c) 18.5	(e) 15.6	(f) 73.5	(g) 5.6	(h) 2.8	(i) 11.6	(j) 75.6	(k) 8.3	(l) 77.2	(m) 5.1	(n) 78.8	(o) 1.1	(p) 80.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	38.1	39.5	44.4	50.3	56.6	61.6	65.4	65.0	59.3	52.9	44.9	39.4
	DBStd	11.68	7.56	7.28	6.53	6.82	6.52	5.96	5.87	5.46	5.20	6.28	6.34	7.32
	HDD50	1501	371	298	194	77	14	0	0	0	0	41	174	332
	HDD65	5131	834	715	639	442	273	137	68	67	185	375	603	793
	CDD50	2060	2	3	20	87	217	349	478	465	281	132	21	5
	CDD65	214	0	0	0	1	12	36	81	67	15	2	0	0
	CDH74	1647	0	0	0	17	88	295	619	537	87	4	0	0
	CDH80	430	0	0	0	0	14	69	173	161	13	0	0	0
Wind	WSAvg	8.2	10.2	9.8	9.1	7.7	7.3	6.9	6.8	6.6	7.2	8.2	8.7	10.0
Precipitation	PrecAvg	30.5	2.7	2.3	2.1	1.9	2.4	2.8	2.9	3.2	2.2	2.4	2.6	3.0
	PrecMax	41.7	5.0	6.0	5.3	4.7	4.6	5.6	6.2	7.9	6.8	6.2	4.4	6.5
	PrecMin	21.7	0.2	0.4	0.2	0.1	0.6	0.9	1.1	0.8	0.3	0.9	0.4	0.6
	PrecStd	4.8	1.3	1.2	1.2	1.0	1.0	1.2	1.2	2.1	1.4	1.1	1.1	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	58.1	66.3	76.5	81.6	87.0	90.2	90.0	81.6	72.8	62.7	56.2
		MCWB	51.4	48.7	52.8	58.8	64.8	68.5	69.1	68.7	65.4	62.2	56.7	52.3
	2%	DB	52.6	53.8	61.1	70.6	76.0	81.1	84.8	84.4	76.1	67.6	58.3	54.0
		MCWB	49.8	48.3	51.5	56.6	61.7	66.5	67.9	66.9	63.9	60.3	54.3	51.0
	5%	DB	50.0	51.3	57.0	65.9	72.3	77.0	80.7	80.2	71.7	64.2	55.9	51.8
		MCWB	47.3	47.4	49.9	54.6	60.2	64.5	66.2	66.0	62.1	58.4	52.7	49.3
	10%	DB	47.9	49.1	53.7	61.5	68.1	73.0	77.0	75.6	68.0	61.5	53.7	49.4
		MCWB	45.5	46.0	48.4	52.3	58.3	62.9	64.7	64.3	60.1	57.1	50.9	47.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	51.3	55.0	61.1	67.1	71.6	71.9	71.8	67.5	63.6	57.4	53.6
		MCDB	54.1	54.7	62.3	73.1	77.4	83.0	84.5	85.2	77.5	70.6	60.9	55.4
	2%	WB	50.0	49.5	52.7	57.8	63.9	68.0	69.3	68.8	65.4	61.1	55.0	51.5
		MCDB	52.0	52.2	58.6	67.1	72.7	77.7	81.1	80.0	73.5	66.1	57.8	53.6
	5%	WB	47.6	48.0	50.9	55.4	61.4	65.8	67.5	66.9	63.2	59.3	53.1	49.6
		MCDB	49.8	50.7	55.9	64.1	69.5	74.9	77.9	77.0	69.8	63.4	55.7	51.7
	10%	WB	45.6	46.1	49.1	53.3	59.2	63.8	65.9	65.2	61.4	57.6	51.1	47.3
		MCDB	47.7	49.0	53.3	60.6	66.8	71.8	74.9	73.8	66.9	61.1	53.4	49.3
Mean Daily Temperature Range	5% DB	MDBR	7.6	9.0	12.0	15.0	15.3	15.9	15.7	15.4	13.7	11.4	8.3	7.3
		MCDBR	9.0	11.5	17.5	21.1	21.3	22.3	21.9	22.8	18.6	14.4	10.2	8.6
		MCWBR	7.8	8.3	10.4	11.0	10.9	11.3	9.3	9.5	9.0	8.6	7.7	7.4
	5% WB	MCDBR	9.0	9.5	14.5	18.6	18.7	19.6	19.5	19.7	16.0	12.9	9.5	8.4
		MCWBR	8.2	7.7	9.5	10.4	10.5	11.0	9.2	9.4	8.6	8.5	7.7	7.6
Clear-Sky Solar Irradiance	taub	0.322	0.344	0.377	0.412	0.422	0.419	0.423	0.414	0.385	0.360	0.335	0.320	
	taud	2.406	2.371	2.282	2.201	2.217	2.243	2.234	2.281	2.349	2.418	2.410	2.373	
	Ebn at Noon	223	248	259	262	265	266	263	259	254	239	215	203	
	Edh at Noon	20	26	34	41	43	42	42	38	32	25	20	18	
All-Sky Solar Radiation	RadAvg	246	459	826	1312	1562	1700	1611	1372	1033	610	290	193	
	RadStd	35	90	117	150	151	151	193	141	107	51	34	29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page